FuseLens: Attention-Guided Gene Fusion Classification via Long-Context Genomic Foundation Models

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December 24, 2025

Abstract

Gene fusions are established oncogenic drivers, yet high-precision classification of fusion candidates remains challenging due to repetitive sequence content, paralogy, and diverse artifact modes produced by RNA-seq fusion pipelines. We introduce FuseLens, a long-context sequence classifier that fine-tunes HyenaDNA [5], a sub-quadratic ($O(n \log n)$) foundation model for long biological sequences, to classify fixed-length windows of $n = 32{,}768$ characters at single-position resolution. Each sample window is constructed by concatenating upstream and downstream sequence segments around a candidate junction and applying a random jitter operator that uniformly places the resulting core inside the 32 kb window via random DNA padding (or random cropping when the core exceeds 32 kb), reducing positional shortcut learning. FuseLens uses a learnable attention pooling head to produce both a fusion probability and an attention-derived breakpoint proxy, reported only under a confidence gate ($\tau = 0.9$). On a curated dataset of 253,925 sequences assembled from multiple positive fusion sources and an explicitly engineered negative space, FuseLens achieves strong blind test performance (Accuracy = 0.9307, AUC = 0.9794, MCC = 0.8513). Post-hoc analysis on true positives ($N = 12{,}737$) reveals that the model learns a variable-resolution localization strategy: while the mean attention width is 5,435 bp, the median width is significantly sharper at 2,095 bp, indicating that FuseLens frequently narrows its focus to the immediate vicinity ($\sim$6%) of the structural breakpoint within the 32 kb window.

1 Introduction

Gene fusions—hybrid gene products created through chromosomal rearrangements or transcriptional mechanisms—are clinically actionable biomarkers in multiple cancer types. Widely used RNA-seq fusion callers such as STAR-Fusion [1] and Arriba [2] rely on short-read alignment signatures (e.g., split reads and discordant pairs). While effective in many settings, alignment-first pipelines may produce false positives in low-complexity regions, paralogous loci, or in the presence of sequencing artifacts, and they do not explicitly learn long-range sequence context around candidate junctions.

Deep learning provides a complementary "sequence-first" approach: instead of hand-engineered evidence rules, a model learns discriminative sequence grammar. Early CNN-based approaches (e.g., FusionAI [3]) demonstrated feasibility but were constrained by limited receptive fields. Transformer-based approaches capture global context but incur quadratic attention cost ($O(n^2)$), making single-position modeling over tens of kilobases computationally expensive.

FuseLens adapts HyenaDNA [5], which replaces quadratic attention with implicit long convolutions and sub-quadratic mixing ($O(n \log n)$) [14], enabling 32 kb processing at high resolution. FuseLens is designed as a **post-caller validation/classification** module: given a candidate junction represented as two flanking sequence segments, FuseLens outputs a fusion probability and a localization proxy derived from attention.

1.1 Contributions

- **Foundation Model Adaptation:** We present FuseLens, a long-context fusion classifier developed by fine-tuning the `hyenadna-small-32k` genomic foundation model [5], enabling sub-quadratic processing of 32 kb windows.

- **Translation Invariance Strategy:** We introduce a Shift-Invariant Junction Operator ($\mathcal{J}$) that enforces robustness to breakpoint placement, preventing the model from relying on positional heuristics (e.g., center bias).

- **Hard-Negative Engineering:** We construct an explicitly engineered negative space—combining canonical sequences, grammar inversions, and reverse-complement augmentations—to rigorously suppress spurious false positives.

- **Weakly Supervised Localization:** We demonstrate that attention geometry provides an effective proxy for breakpoint localization (AUC=0.9794), allowing for coarse-grained junction discovery without the need for dense, nucleotide-level supervision.

2 Biological Background: Chimeric Transcripts in Cancer

Chimeric transcripts can arise through multiple mechanisms [6].

2.1 Mechanisms of Formation

- **Read-Through (Cis-SAGe):** transcriptional read-through between adjacent genes without a DNA rearrangement.

- **Chromosomal Rearrangement:** inversions (e.g., *EML4–ALK*) and inter-chromosomal translocations (e.g., *BCR–ABL1*).

- **Cis-Type SV-Driven Fusion:** deletions/duplications that bring two loci into contiguity.

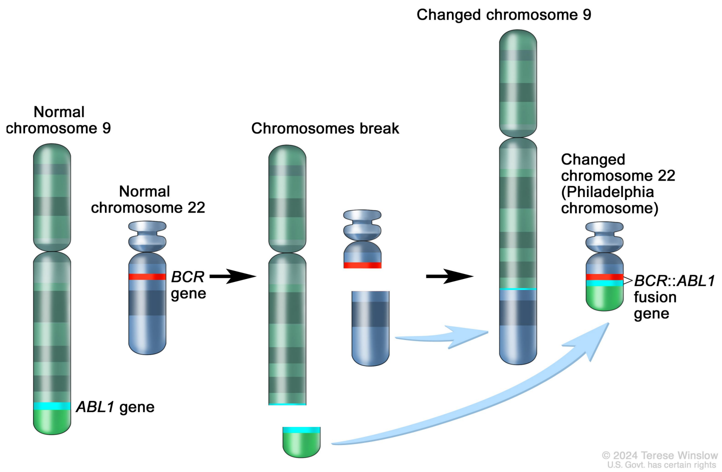

Figure 1: **The BCR–ABL1 fusion event.** Reciprocal translocation t(9;22) creates a fusion gene on derivative chromosome 22, characteristic of CML [7].

2.2 Examples for Key Oncogenic Fusions

We focus on detecting high-impact fusions characterized in the literature such as:

- **BCR-ABL1 (CML):** Resulting from the t(9;22) translocation (Philadelphia Chromosome). It creates a constitutively active tyrosine kinase driving Chronic Myeloid Leukemia [7].
- **TMPRSS2-ERG (Prostate Cancer):** A fusion of the androgen-responsive promoter of *TMPRSS2* with the oncogene *ERG*. Found in ∼50% of prostate cancers [8].
- **SLC45A3-ELK4 (Prostate Cancer):** A prominent example of **Cis-Splicing**. This chimera can form via transcriptional read-through without any underlying DNA mutation [9].

2.3 Non-Cancerous Physiological Chimeras

Not all chimeric transcripts are pathogenic. Normal physiological chimeras occur in healthy cells as passengers of transcription and can drive evolutionary adaptation or protein diversity.

- **Benign Read-Throughs:** Transcripts like *ZFAND3-ZFAND6* occur in normal human tissues via Cis-SAGe (Splicing of Adjacent Genes) to create novel protein isoforms without genomic rearrangement [10].
- **Normal Trans-Splicing:** The *JAZF1-JJAZ1* RNA chimera, originally thought to be specific to endometrial sarcoma, has been identified in normal endometrial stroma cells resulting from trans-splicing rather than DNA fusion [11].
- **Evolutionary Adaptation:** Chimerism is a mechanism of gene birth. The *Jingwei* gene in *Drosophila* originated ∼2.5 million years ago from the retrotransposition of *Adh* into the *Ymp* gene, conferring new metabolic capabilities [12].

Distinguishing these benign physiological events from oncogenic drivers is a critical challenge for high-precision clinical classifiers.

3 Related Work

- **Alignment-Based Detectors:** Tools like STAR-Fusion [1] and Arriba [2] define the current clinical standard but are prone to high False Positive rates in complex genomic regions.
- **Deep Learning in Genomics:** Models like FusionAI [3] demonstrated the feasibility of using neural networks for structural variant detection.
- **Genomic Foundation Models:** HyenaDNA [5] and Nucleotide Transformer [13] represent a shift towards 'pre-training + fine-tuning."

4 Problem Formulation and Scope

FuseLens addresses **fusion-candidate window classification**. The input is not an entire genome scan; rather, it is a variable-length sequence window candidate (e.g., fusion caller output, database curation, or structural variant hypotheses).

4.1 Inputs

Each sample s is a variable-length sequence window

$$s \in \Sigma^l, \quad l \leq 32{,}768 \tag{1}$$

where Σ is the character alphabet $\{A, C, G, T\}$ used by the tokenizer.

4.2 Outputs

FuseLens returns:

- A fusion probability $\hat{p} = P(y = 1 \mid s)$
- A localization proxy $\hat{\ell}$ derived from attention, only emitted if $\hat{p} \geq \tau$

4.3 Labels

Each window receives a binary label $y \in \{0, 1\}$:

- $y = 1$: fusion-positive windows (database and high-confidence fusion sets).
- $y = 0$: negative fusion, described in Section 6.

5 Model Architecture and Methods

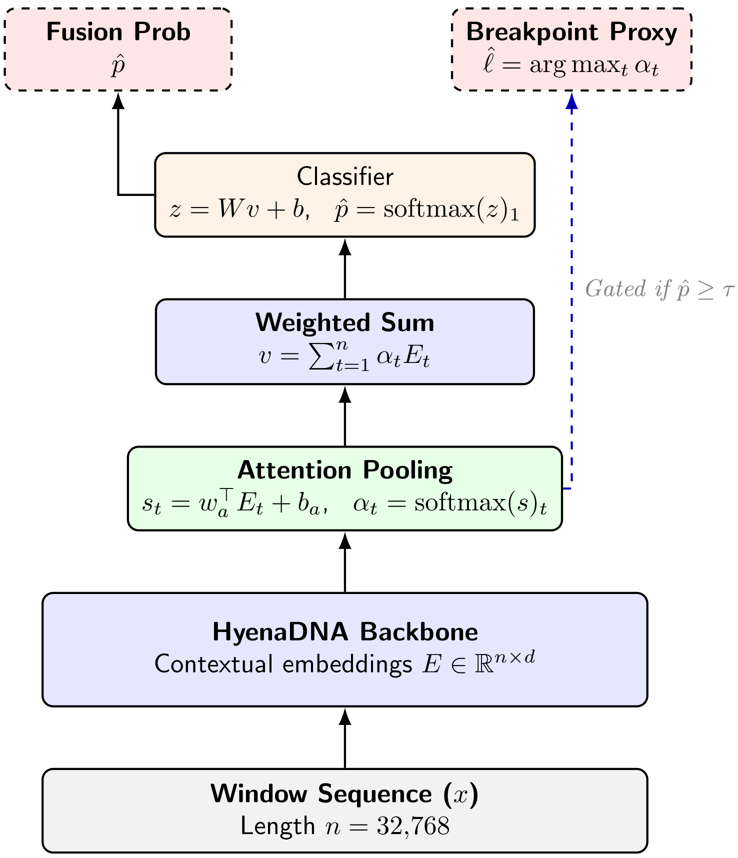

Figure 2: **FuseLens architecture.** HyenaDNA produces contextual token embeddings, attention pooling aggregates evidence for classification, and the attention peak provides a localization proxy under confidence gating.

5.1 Window construction with random jitter

To reduce positional shortcuts (e.g., "junction always near the center"), we apply padding with **random jitter** to map s into a fixed-length window $x \in \{A, C, G, T\}^n$, $n = 32,768$:

$$\text{Jitter}(s) = \begin{cases} r_\ell \oplus s \oplus r_r & \text{if } m < n \\ s[k : k+n] & \text{if } m \geq n \end{cases} \tag{2}$$

where $|r_\ell| + |r_r| = n - m$ and $|r_\ell| \sim \text{Unif}\{0, \dots, n-m\}$ (uniform random placement), while $k \sim \text{Unif}\{0, \dots, m-n\}$ (random crop). Padding strings r_ℓ, r_r are sampled IID from $\{A, C, G, T\}$.

5.2 HyenaDNA backbone

FuseLens fine-tunes the pretrained HyenaDNA model [5]. Hyena-style operators approximate attention-like global mixing using implicit long convolutions [14]:

$$\text{Hyena}(u) = (H_1(u) * \psi) \odot H_2(u) \tag{3}$$

where $u \in \mathbb{R}^{n \times d}$ and d is the model dimension for the embedded input, ψ is a learned long filter, $*$ is convolution along sequence length, and $\odot$ is element-wise gating. The output is contextual embeddings $E \in \mathbb{R}^{n \times d}$.

5.3 Attention pooling and classification

First, we compute an unnormalized scalar score (logit) s_t for each nucleotide position t via a learnable **linear projection**:

$$s_t = w_a^\top E_t + b_a \tag{4}$$

To derive the final attention weights α_t, we apply the softmax function over the sequence length n, normalizing the logits into a probability distribution:

$$\alpha_t = \frac{e^{s_t}}{\sum_{j=1}^{n} e^{s_j}} \tag{5}$$

The final pooled sequence representation $v \in \mathbb{R}^d$ is the weighted sum:

$$v = \sum_{t=1}^{n} \alpha_t E_t \tag{6}$$

Finally, the classifier produces logits $z \in \mathbb{R}^2$ and fusion probability:

$$z = Wv + b, \qquad \hat{p} = \text{softmax}(z)_1 \tag{7}$$

5.4 Confidence-gated weak localization

We define the attention-peak location:

$$\hat{\ell} = \arg \max_{t \in \{1, \dots, n\}} \alpha_t \tag{8}$$

We only report localization if $\hat{p} \geq \tau$:

$$\hat{\ell}_{\text{final}} = \begin{cases} \hat{\ell} & \text{if } \hat{p} \geq \tau \\ \varnothing & \text{otherwise} \end{cases} \tag{9}$$

This suppresses spurious localization for low-confidence predictions.

5.5 Attention-geometry: half-maximum breakpoint region

For analysis (not training), we define a half-maximum attention region:

$$\mathcal{R} = \{t : \alpha_t \geq 0.5 \cdot \max_j \alpha_j\} \tag{10}$$

with width

$$\text{width}(\mathcal{R}) = \max(\mathcal{R}) - \min(\mathcal{R}) \tag{11}$$

This measures whether attention behaves like a sharp pointer or a broad regional evidence map.

6 Negative Data

A central goal is to construct negatives that reduce shortcut learning and approximate difficult false-positive scenarios.

6.1 Negative sources

The negative pool used in training comprises:

- A labeled **negative CSV** set (20,000 sequences from FusionAI training data).
- A **canonical negative FASTA** set (15,000 sequences) derived from human canonical coding sequences (CDS).
- A **synthetic hard-negative FASTA** set (45,000 sequences) produced by applying structured operators (reverse complement, reversal, and circular permutation) to the canonical sequences.

6.2 Hard-negative operators

Given a canonical sequence s, We define a set of canonical sequences $S = \{s_1, ..., s_N\}$ where $s_i \in \{A, C, G, T\}^L$ derived from UniProt Swiss-Prot [15] and apply three mathematical operators to generate "Hard Negatives". Together, these operators progressively eliminate trivial discriminative shortcuts.

1. **Reverse Complement Operator ($\mathcal{RC}$):**
   Let $\mathcal{C}$ be the base-pairing function ($A \leftrightarrow T, C \leftrightarrow G$).

   $$\mathcal{RC}(s_i) = (\mathcal{C}(b_L), \mathcal{C}(b_{L-1}), \ldots, \mathcal{C}(b_1)) \tag{12}$$

   This generates the sequence of the opposing DNA strand. Since most genomic features (e.g., splice sites, promoter motifs) are strand-specific, this operator forces the model to learn the strict $5' \to 3'$ syntax of the coding strand.

2. **Grammar Inversion Operator ($\mathcal{R}_{seq}$):**
   To generate decoys that preserve nucleotide composition but destroy biological syntax, we apply a strict sequence reversal *without* complementation:

   $$\mathcal{R}_{seq}(s_i) = (b_L, b_{L-1}, \ldots, b_1) \tag{13}$$

   Unlike random shuffling (which destroys local k-mer statistics), $\mathcal{R}_{seq}$ preserves local dinucleotide frequencies while inverting the global grammatical structure (e.g., Start/Stop codon ordering). This presents a harder adversarial challenge than simple permutation.

3. **Shift-Invariant Junction Operator ($\mathcal{J}$):**
   A naive approach to creating negative junctions is to join two random genes ($s_i \oplus s_j$). However, this structurally constitutes a valid fusion event, creating false labels. Instead, we implement a *Circular Permutation* strategy on a single gene s_i:

   $$\mathcal{J}(s_i, k) = s_i[k : L] \oplus s_i[1 : k] \tag{14}$$

   where $k \sim \text{Unif}(0, L)$ and $\oplus$ denotes concatenation. This operator creates a "decoy junction" (a discontinuity) and forces the model to search the entire **32kb** window for anomalies, but ensures the resulting sequence remains monogenic (Label 0). By sampling k uniformly, we eliminate the "Center Bias" ($L/2$) assumption found in previous literature.

Note: While UniProt Swiss-Prot entries are curated at the protein level, we derived our training data from the canonical coding sequences (CDS) corresponding to these validated entries [15], ensuring nucleotide-level compatibility with the model.

6.3 Dataset-level reverse-complement augmentation

In the training pipeline, the raw negative pool is augmented by adding reverse complements.

7 Dataset Composition and Statistics

FuseLens was trained and evaluated on a comprehensive dataset of **253,925** unique genomic windows:

- **Positives: 93,925.** Aggregated from a labeled positive CSV (30,745) and three high-confidence FASTA sources: BLAST-validated chimeras (6,435), COSMIC verified sequences (13,279), and an additional chimera dataset (43,466). Length statistics indicate a mean of 7,905 bp (min: 101 bp, max: 34,308 bp). Only 1 sequence exceeded the 32 kb limit and was randomly cropped, while the remaining 93,924 were padded via the random jitter strategy.

- **Negatives: 160,000.** Constructed as described in 6. This base pool was explicitly doubled via reverse-complement augmentation to enforce strand invariance.

As detailed in Table 1, the dataset retains a robust class balance ($\sim$37% Positive vs. 63% Negative), deliberately over-sampling negatives to penalize false-positive predictions in clinical scenarios.

Table 1: Final Dataset Distribution and Splits

Split	**Total Samples**	**Positives**	**Negatives**
Training	177,740	65,745	111,995
Validation	38,096	14,091	24,005
Testing	38,089	14,089	24,000
Total Pool	**253,925**	**93,925**	**160,000**

8 Results

8.1 Blind test performance

Metric	Score
Accuracy	0.9307
Precision	0.9083
Recall	0.9040
F1 Score	0.9062
AUC	0.9794
MCC	0.8513

Table 2: Blind test metrics.

8.2 Discriminative Performance and Robustness

FuseLens demonstrates strong discriminative capacity on the blind test set, achieving an Area Under the Curve (AUC) of **0.9794**. As shown in Figure 3, the model effectively separates true fusions from the engineered negative space, though the transition region indicates a non-trivial decision boundary.

Key observations from the inference dynamics (Figure 3) include:

- **Confusion Matrix (Panel A):** The model correctly identified 12,737 of 14,089 fusion samples (Recall 90.40%) and 22,714 of 24,000 negatives (Specificity 94.6%). The imbalance between False Positives (1,286) and False Negatives (1,352) is symmetric, suggesting no strong bias toward either class despite the class-imbalanced training data.

- **Confidence Distribution (Panel C):** The probability distribution is bimodal, with the majority of negatives clustered near $p = 0.0$ and positives near $p = 1.0$. However, unlike simpler baselines, we observe a "soft tail" of samples in the 0.8-0.95 range (green curve), indicating that certain fusion isoforms present a harder classification challenge, potentially due to sequence homology with the negative decoys.

- **Predicted Breakpoint Distribution (Panel D):** The distribution of predicted attention peaks spans the entire window, confirming that the model does not suffer from a centralized bias. However, a notable accumulation of peaks is observed near index 0, suggesting a "default" behavior where the model attends to the sequence start when evidence is ambiguous.

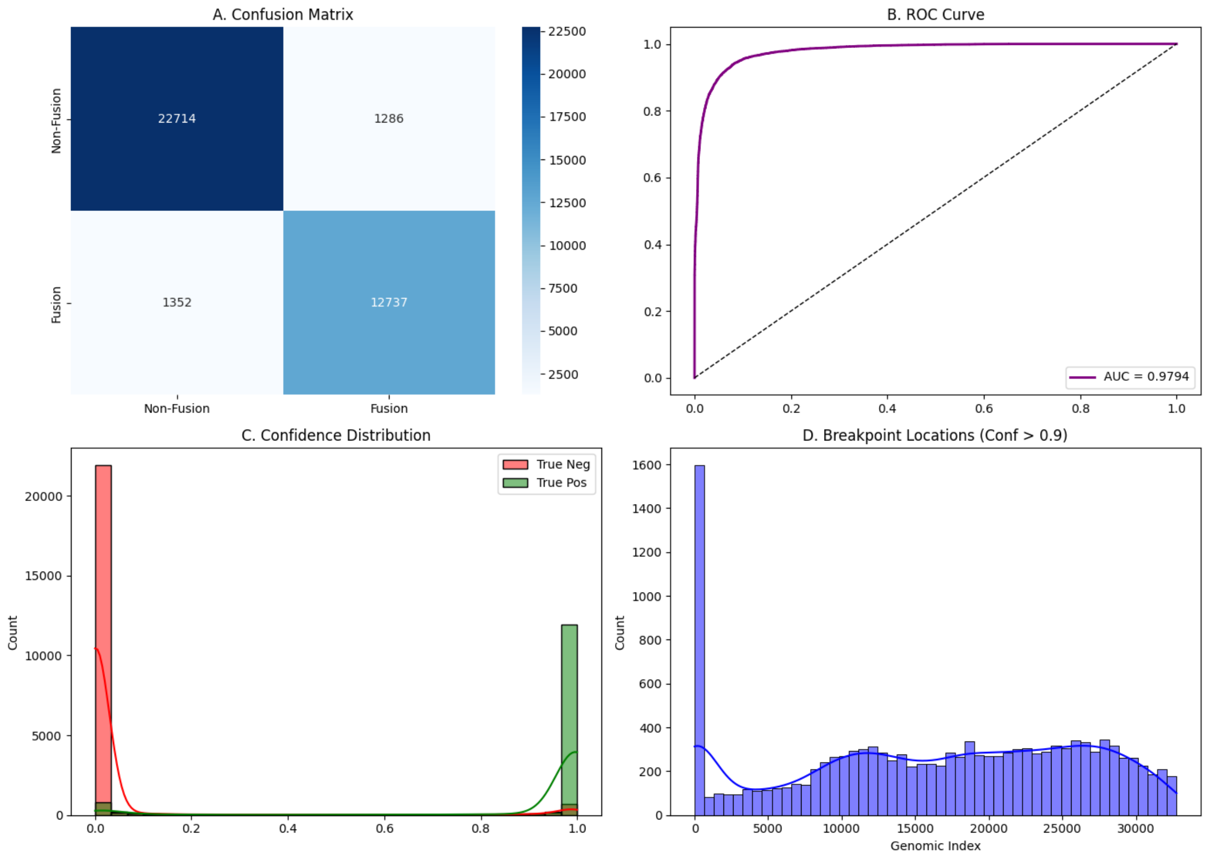

Figure 3: **Inference dynamics on blind test set.** (A) Confusion matrix showing strong diagonal dominance. (B) ROC curve (AUC=0.9794). (C) Confidence distribution showing bimodal separation with a tail of hard positives. (D) Distribution of predicted attention-peak indices, showing broad coverage with a bias toward the sequence terminus.

8.3 Localization Resolution and Attention Geometry

A critical novelty of FuseLens is its ability to localize breakpoints without explicit segmentation labels. We analyzed the "Half-Maximum Width" of the attention distributions for all True Positives ($N = 12,737$).

- **Median Width:** 2,095 bp (~6% of the 32 kb window).
- **Mean Width:** 5,435 bp.

This discrepancy between the median and mean indicates a **variable-resolution strategy**. As visualized in Figure 4, the model's behavior falls into distinct regimes:

1. **Sharp Localization (e.g., Sample #2):** In high-confidence cases, the attention mechanism collapses into a narrow peak (e.g., width 264 bp), effectively pinpointing the junction.
2. **Regional Aggregation (e.g., Sample #3):** For more complex or repetitive sequences, the model attends to a broad region (width $>$ 10 kb), aggregating evidence across the entire gene body rather than committing to a single breakpoint.

This suggests that FuseLens acts as a "Soft Pointer": it offers high-resolution localization when the signal is unambiguous, but gracefully degrades to regional detection when the sequence context requires broader integration.

Figure 4: **Heterogeneity in attention geometry.** (Top) Sample #1: Edge artifact peak. (Middle) Sample #2: Sharp localization (264 bp width) indicating precise junction discovery. (Bottom) Sample #3: Broad regional attention (13,718 bp width) indicative of evidence aggregation over large genomic spans.

9 Training Configuration and Reproducibility

FuseLens was trained with:

- Backbone: `LongSafari/hyenadna-small-32k-seqlen-hf` [5]
- Window length: 32,768
- Batch size: 8
- Epochs: 3
- Optimizer: AdamW with learning rate 1×10^{-5}
- Schedule: linear schedule with 100 warmup steps (transformers `get_linear_schedule_with_warmup`)

Hardware: NVIDIA A100 80GB GPU.

Reproducibility Note: A Git Repository is available here: https://github.com/UdiRuni/FuseLens and includes all the data construction, augmentation operators, model architecture, and evaluation scripts used to generate the reported metrics and attention-geometry statistics.

10 Discussion

10.1 Why long context helps

Gene fusion signals may be distributed across exons, splice motifs, and partner-specific sequence context. HyenaDNA's computes long convolutions using FFT that allows the classifier to integrate evidence across up to 1M kb windows without quadratic self-attention cost [14].

10.2 Random jitter reduces positional shortcuts

A major failure mode in junction modeling is a fixed positional prior (e.g., "junction near the middle"). The random jitter operator uniformly randomizes the placement of the core signal within each 32 kb window, encouraging translation-invariant feature learning.

10.3 Variable Granularity in Localization

Our updated analysis reveals that FuseLens does not strictly behave as a broad regional aggregator. The significantly lower median width (2,095 bp) compared to the mean (5,435 bp) suggests that for a large subset of fusions, the model is capable of quasi-segmentation behavior. This sharpness likely results from the improved negative engineering (using Ensembl CDS), which forces the model to distinguish fusion breakpoints from valid splice sites, requiring finer-grained attention than distinguishing DNA from random noise.

11 Limitations and Future Work

- **Start-token Bias:** As seen in Figure 3D, there is a disproportionate accumulation of attention peaks at the start of the sequence window (Index 0). This serves as a "default" attention sink for difficult or ambiguous samples. Future work could introduce a "No-Op" token to absorb this probability mass explicitly.
- **Random DNA padding realism.** Random A/C/G/T padding enforces translation invariance but may not reflect real genomic flanks. Future work should test padding with real reference flanks or composition-matched backgrounds.
- **Attention-based localization is approximate.** Attention peaks are not guaranteed to coincide with true breakpoint coordinates. Future work should validate localization against experimentally confirmed breakpoints (e.g., long-read sequencing).
- **Baselines and ablations are needed.** To attribute gains to long context vs negative engineering, we want to add: (i) CNN baseline, (ii) short-context truncation baseline, (iii) ablations removing each negative operator and attention pooling.

12 Future directions

1. **Clinical integration as a validation layer.** FuseLens can be positioned after candidate generation (e.g., STAR-Fusion/Arriba outputs) as a high-precision classifier to reduce false positives.
2. **Synergy with Long-Read Sequencing.**
   The 32 kb receptive field of FuseLens is inherently compatible with third-generation sequencing technologies like Oxford Nanopore. This aligns with tools such as CTAT-LR-fusion [17], which leverage long reads to identify complex isoforms without assembly. By operating as a sequence-level neural validator, FuseLens can complement alignment-based callers like CTAT-LR-fusion.
3. **Hard-negative realism.** Replace heuristic decoys with artifact-aware negatives derived from real RNA-seq debris and known failure modes (repeats, paralogs, read-throughs).
4. **Improved localization objectives.** Add supervised or semi-supervised auxiliary losses (pointer/segmentation) when breakpoint coordinates are available.

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